CHAPTER 4 / SUPPLEMENTARY MATERIAL

Cyber–Physical Security and Mission Assurance for Unmanned Aerial Vehicles

UAS Data Links: Functions, Attributes, Latency, and Security Constraints

Companion figures, service requirements, a latency-budget example, and a communication-assessment exercise

Learning goal: define what each link must deliver and verify its behavior under degraded conditions.

Onboard networks and external data links

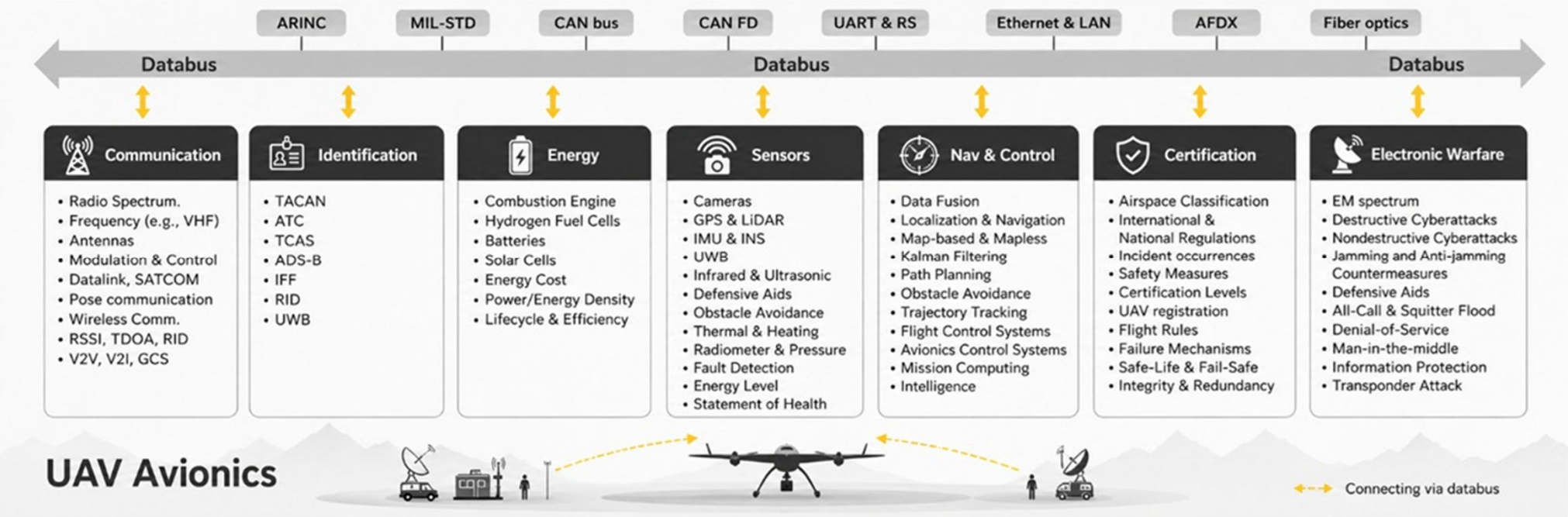


Figure S4.1. Internal buses connect onboard components. External links carry services between aircraft, operators, and other systems. This avionics map is illustrative.

Conceptual illustration. Applicable requirements and performance depend on the selected UAS implementation.

Secure data-link design principles

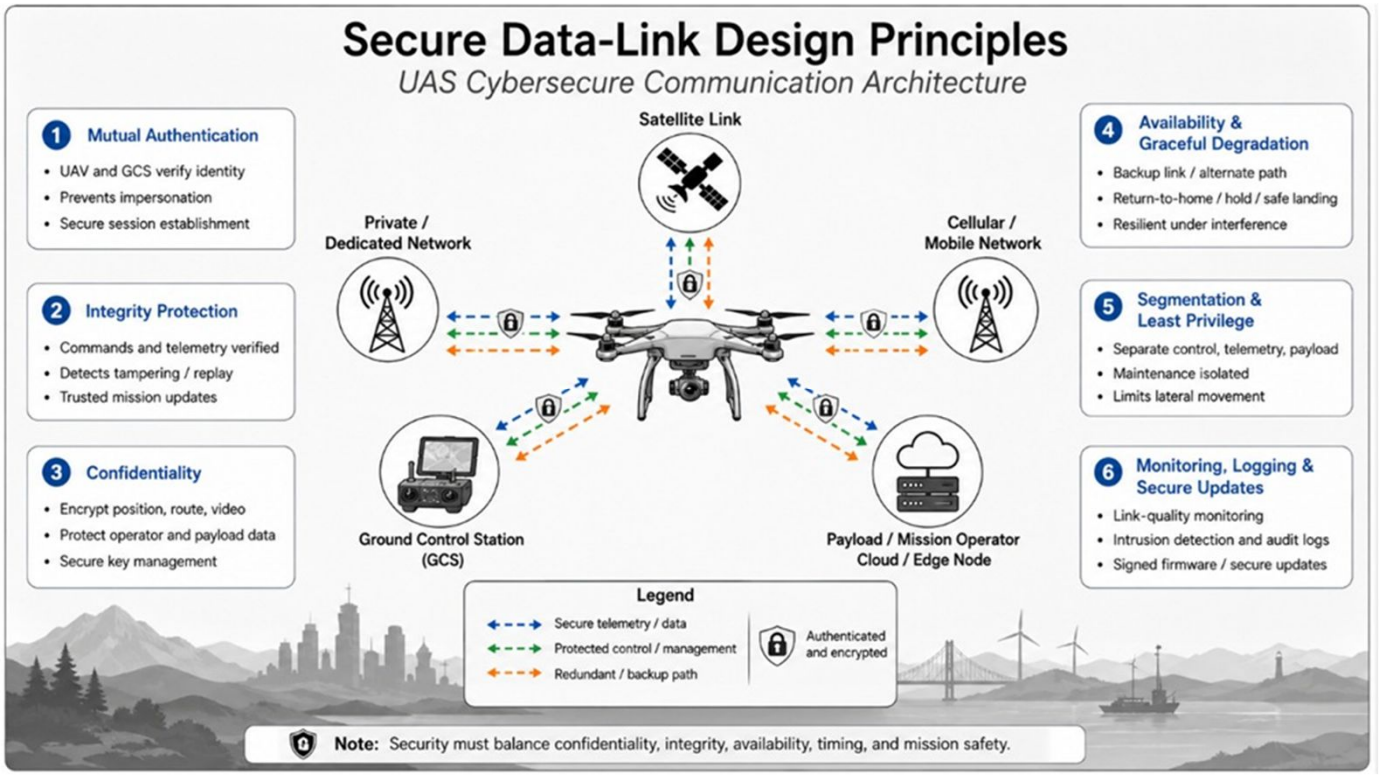


Figure S4.2. Authentication, integrity, confidentiality, and availability address different requirements. Message freshness and recovery behavior need explicit design choices.

Conceptual illustration. Applicable requirements and performance depend on the selected UAS implementation.

Data-link threats and mission consequences

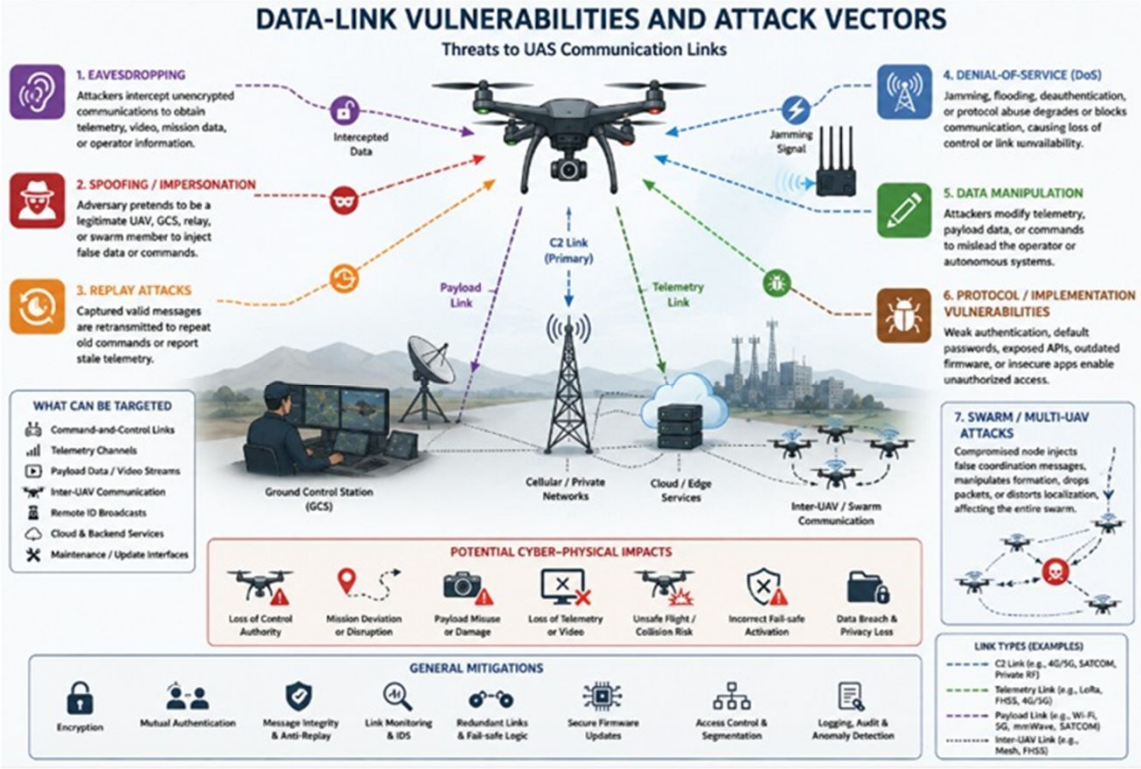


Figure S4.3. Exposed interfaces can support interception, impersonation, replay, or disruption. The consequence depends on access, implementation, and the receiving application.

Conceptual illustration. Applicable requirements and performance depend on the selected UAS implementation.

Communication resilience and recovery

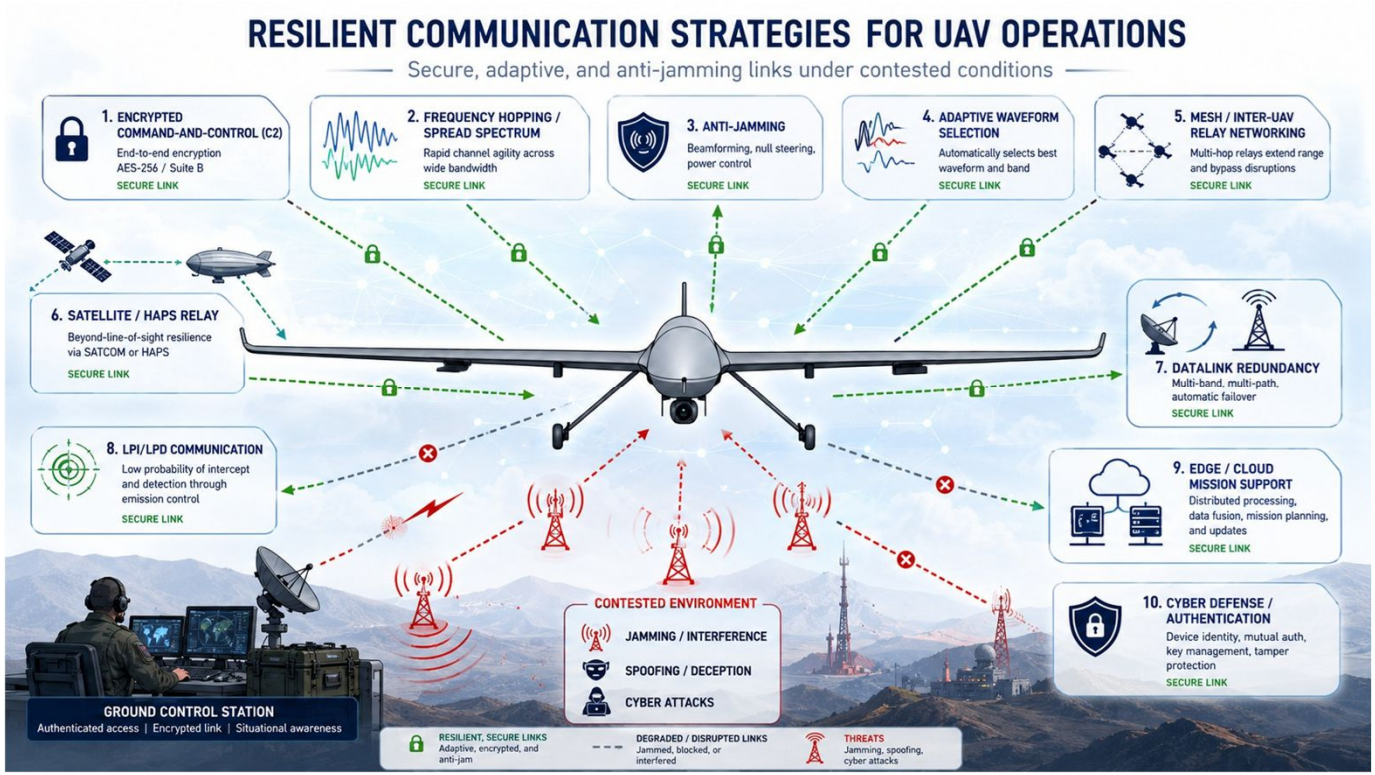


Figure S4.4. Diversity, adaptive links, and local fallback can improve resilience within defined conditions. Shared dependencies and switching delays can limit their benefit.

Conceptual illustration. Applicable requirements and performance depend on the selected UAS implementation.

Link services and their requirements

A shared radio can carry several services. Each service still needs its own timing and failure policy.

Service	Requirement to define	Behavior to validate
Command and control	Authorized commands, freshness, and an application deadline	Reject stale or invalid commands and enter a defined fallback
Telemetry	Update interval, age limit, and critical-state visibility	Flag stale data and distinguish missing data from unchanged state
Payload stream	Useful throughput, confidentiality, and acceptable quality loss	Reduce payload demand before it starves critical traffic
Inter-UAV coordination	Peer identity, shared-state age, and consistency requirements	Handle missing peers and reject obsolete state updates
Maintenance and updates	Authorized access, update integrity, and permitted operating modes	Prevent unauthorized changes and verify recovery behavior

Define priorities and limits at the application level, then verify them under shared-load conditions.

Worked example: sensing-to-actuation budget

Illustrative assumptions: 50 Hz sampling, a 125-byte frame, a 1 Mbit/s link, and a 60 ms deadline.

Budget component	Assumed time (ms)	Interpretation
Sampling wait	20	Maximum wait within one sampling interval
Queueing / scheduling	8	Wait before channel access or transmission
Frame serialization	1	125 × 8 bits ÷ 1,000,000 bit/s
Transport and forwarding	12	Additional network / relay delay
Security and application checks	4	Verification, freshness, and authorization
Actuation processing	5	Application of the accepted command

Total = 20 + 8 + 1 + 12 + 4 + 5 = 50 ms

The assumed 60 ms deadline leaves 10 ms of margin. Retries, bursts, and handovers can consume it.

Security controls and engineering trade-offs

Control selection should preserve message validity, timely delivery, and a defined response to failure.

Control	Benefit and limitation	Verification question
Authenticated messaging and replay protection	Protect origin and integrity. Freshness still needs explicit state.	Do duplicates, stale sessions, and out-of-order messages behave correctly?
Segmentation and traffic scheduling	Restrict exposure and contention. Shared compute can remain a bottleneck.	Can a payload or maintenance load delay critical command processing?
Redundant links and handover	Provide alternate paths. Common failures can affect both.	Does switching preserve authority and meet the interruption budget?
Monitoring and local fallback	Detect degradation and bound impact. Thresholds can cause false alarms.	Do mode transitions remain safe when link or navigation data are stale?

A secure session can still miss a deadline. A backup path can still carry stale or invalid state.

Practical exercise: communication assessment

Compare a baseline, increased payload traffic, a link outage, and recovery in a repeatable simulation or bench test.

1. Service contract	Choose one critical message. Define its size, rate, deadline, freshness limit, and fallback.
2. Measurement record	Log sequence IDs, timestamps, acceptance decisions, link state, and applied commands.
3. Performance analysis	Report delivery ratio, delay percentiles, deadline misses, outage duration, and recovery time.
4. Review and evidence	Separate network receipt from timely application. Explain failures and residual uncertainty.

Deliverable: a service specification, event log, and baseline-versus-degraded comparison.

References and terminology

Primary references provide general measurement and security guidance for the teaching examples.

RFC 7679 A One-Way Delay Metric for IP Performance Metrics (2016).
www.rfc-editor.org/rfc/rfc7679.html

RFC 4303 IP Encapsulating Security Payload (ESP) (2005).
www.rfc-editor.org/rfc/rfc4303.html

RFC 3550 RTP: A Transport Protocol for Real-Time Applications (2003).
www.rfc-editor.org/rfc/rfc3550.html

NIST SP 800-82 Rev. 3, Guide to Operational Technology Security (2023).
csrc.nist.gov/pubs/sp/800/82/r3/final

C2: command and control GCS: ground control station PDR: packet delivery ratio
RTT: round-trip time HAPS: high-altitude platform station